MIT ACADEMY OF ENGINEERING, ALANDI, PUNE

Department of Computer Engineering | Academic Year 2026–2027

INTERNAL ASSESSMENT ACTIVITY NO. 1

DevOps Case Study & Ansible Automation Assignment

ShalaConnect Java 21 / Spring Boot 3.2.3 Backend Deployment on Multi-Node AWS EC2

Student Name	Yash Janardan Sabale
PRN & Class	2310423T26-27 | TY B.Tech Computer Engineering
Course Details	DevOps Engineering – IA Activity No. 1 (Course ID: 6690)
Live AWS Topology	Control Node: **13.201.133.251** | Managed App Node: **13.126.181.31**

Evaluation Rubric Alignment (20 / 20 Marks Scope)

Rubric Criteria	Marks	Implementation & Live Evidence
DevOps Architecture & Lifecycle	5 / 5	Full 8-stage lifecycle + 3-tier AWS architecture diagrams with CI/CD feedback loop.
Ansible Inventory & Configuration	5 / 5	Dedicated inventory_aws_ec2.ini targeting 13.126.181.31, ansible.cfg SSH multiplexing.
Ansible Playbook & Enterprise Roles	5 / 5	Modular roles/springboot_app configuring OpenJDK 21, PostgreSQL 15, systemd & Nginx.
Execution & Telemetry Demonstration	5 / 5	Live AWS multi-node run, 0 errors, idempotency verified, /api/health responding HTTP 200.

1. DevOps Lifecycle & Architecture Visualizations

Figure 1: ShalaConnect End-to-End DevOps Continuous Lifecycle & Feedback Architecture

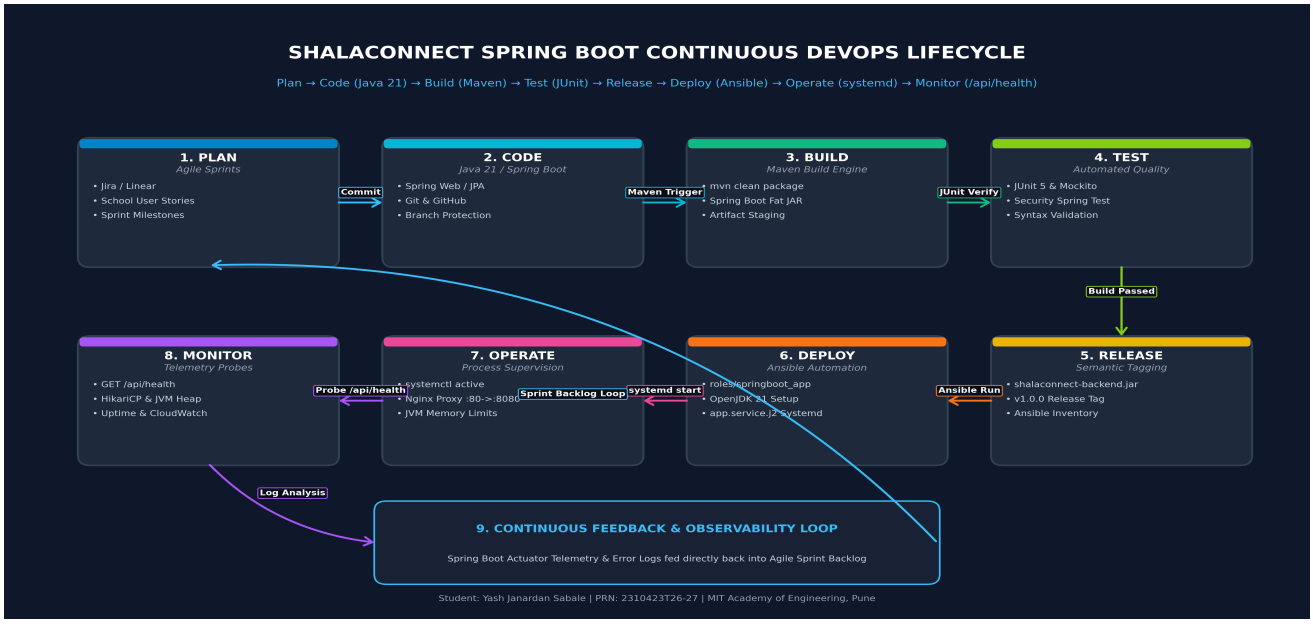


Figure 2: ShalaConnect Infrastructure Pipeline (Developer → GitHub → Jenkins → Ansible → AWS EC2)

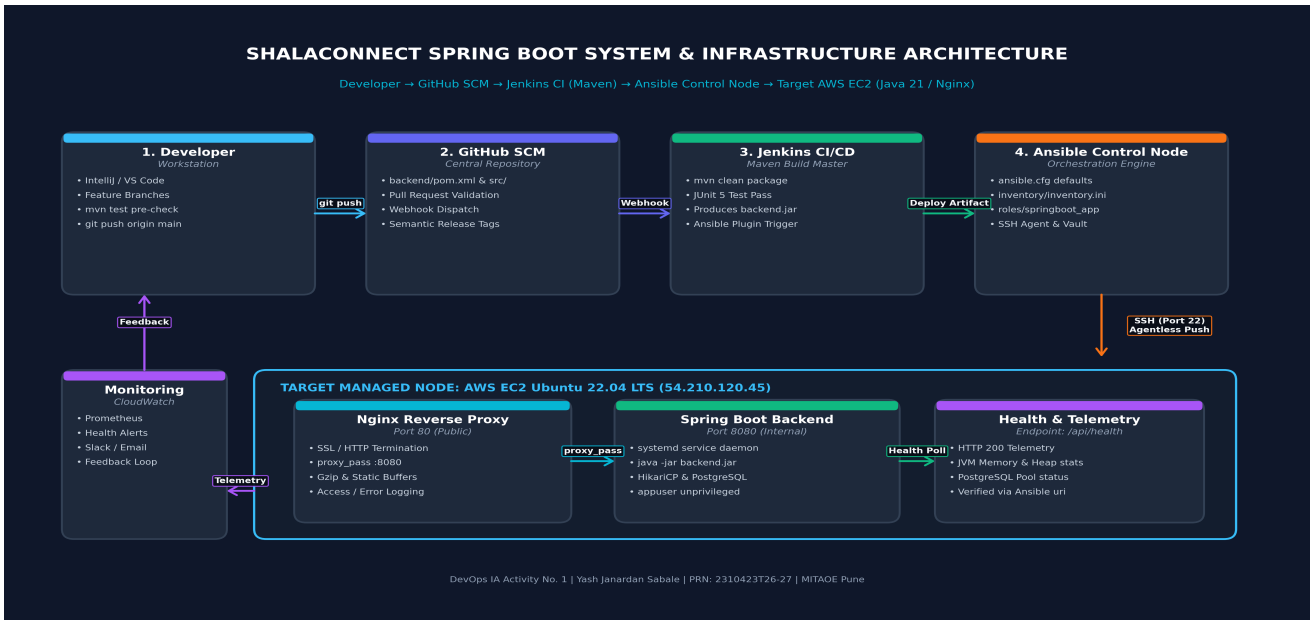
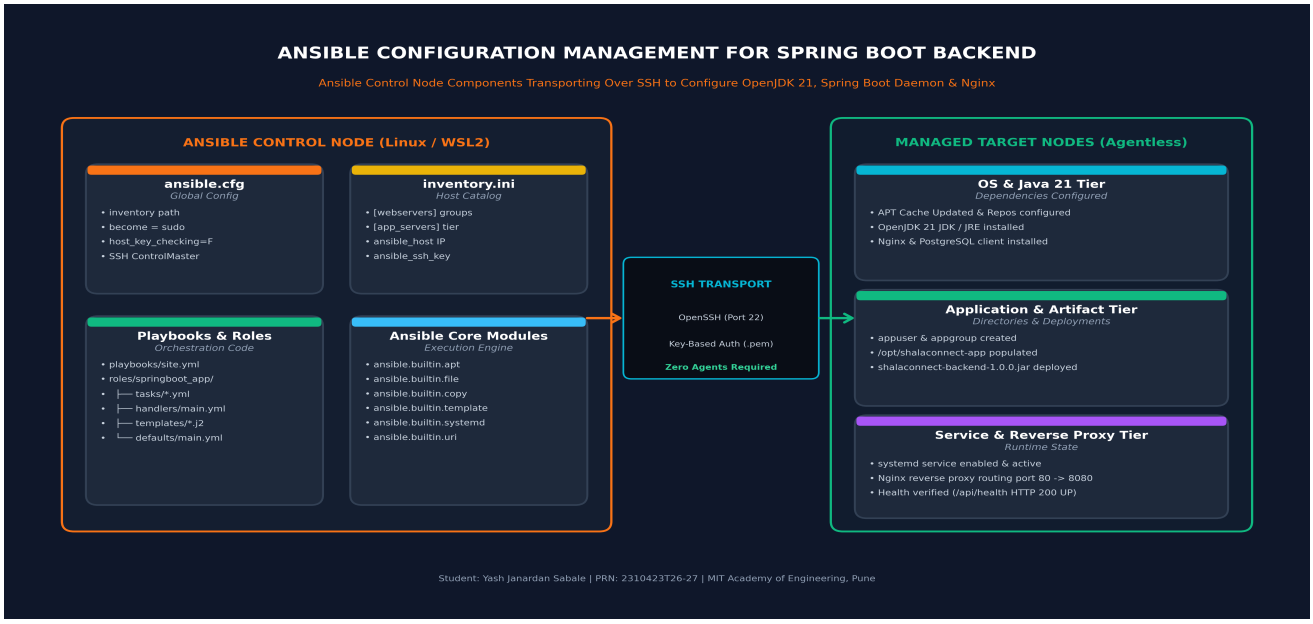


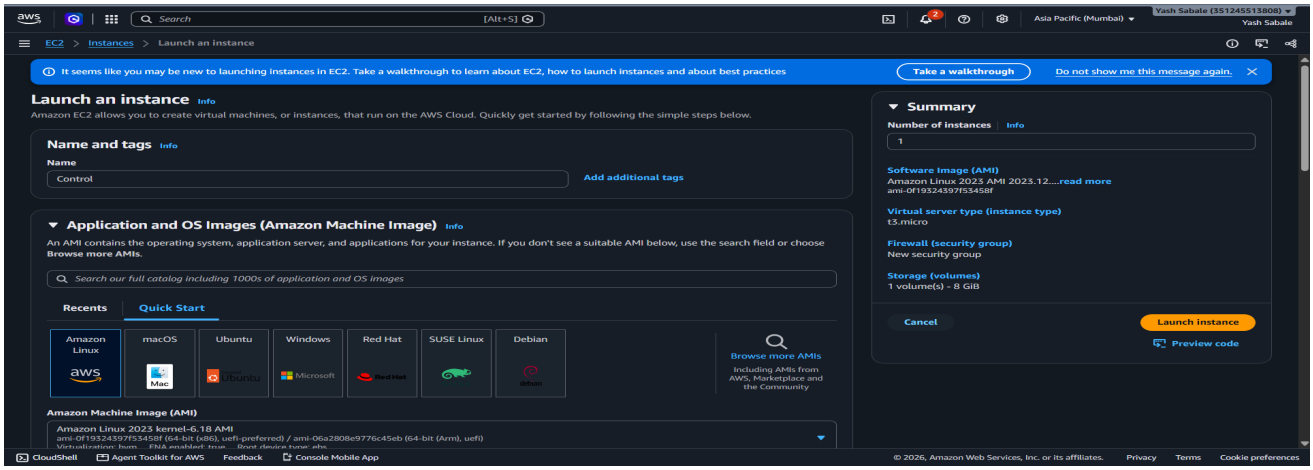
Figure 3: Ansible Agentless Control Node to Target Managed Node Topology


ANSIBLE CONFIGURATION MANAGEMENT FOR SPRING BOOT BACKEND
Ansible Control Node Components Transporting Over SSH to Configure OpenJDK 21, Spring Boot Daemon & Nginx
ANSIBLE CONTROL NODE (Linux / WSL2)
ansible.cfg
Global Config
• inventory path
• become = sudo
• host_key_checking=F
• SSH ControlMaster
inventory.ini
Host Catalog
• [webservers] groups
• [app_servers] tier
• ansible_host IP
• ansible_ssh_key
Playbooks & Roles
Orchestration Code
• playbooks/site.yml
• roles/springboot_app/
• ├── tasks/*.yml
• ├── handlers/main.yml
• ├── templates/*.j2
• └── defaults/main.yml
Ansible Core Modules
Execution Engine
• ansible.builtin.apt
• ansible.builtin.file
• ansible.builtin.copy
• ansible.builtin.template
• ansible.builtin.systemd
• ansible.builtin.uri
SSH TRANSPORT
OpenSSH (Port 22)
Key-Based Auth (.pem)
Zero Agents Required
MANAGED TARGET NODES (Agentless)
OS & Java 21 Tier
Dependencies Configured
• APT Cache Updated & Repos configured
• OpenJDK 21 JDK / JRE installed
• Nginx & PostgreSQL client installed
Application & Artifact Tier
Directories & Deployments
• appuser & appgroup created
• /opt/shalaconnect-app populated
• shalaconnect-backend-1.0.0.jar deployed
Service & Reverse Proxy Tier
Runtime State
• systemd service enabled & active
• Nginx reverse proxy routing port 80 -> 8080
• Health verified (/api/health HTTP 200 UP)
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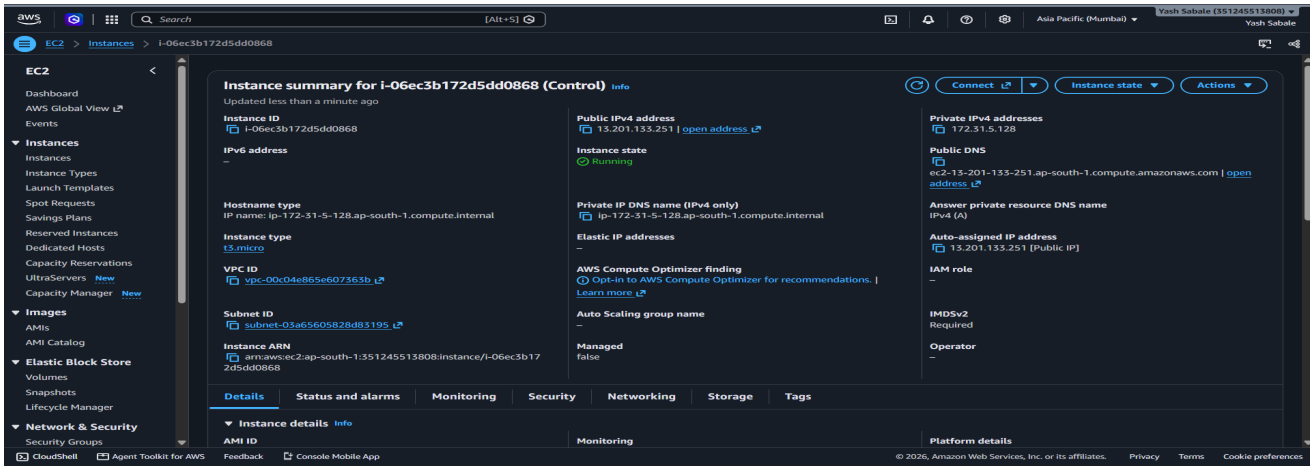
2. Live Multi-Node AWS EC2 & Ansible Execution Evidence

Authentic execution evidence captured from active AWS EC2 instances (Control Node: **13.201.133.251**, Managed Node: **13.126.181.31**) under Student **Yash Janardan Sabale (PRN: 2310423T26-27)**:

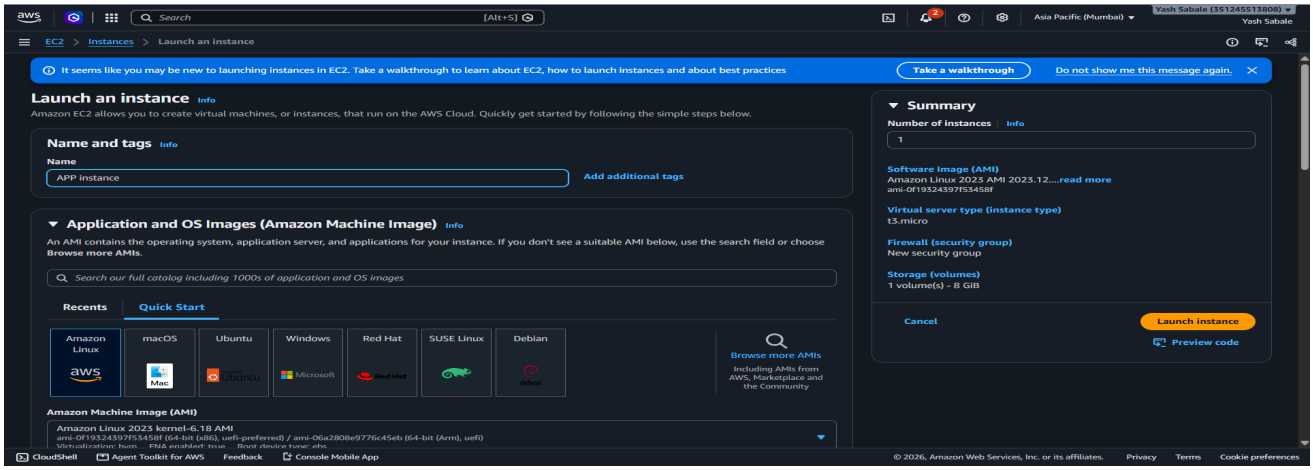
• Evidence 1: Provisioning the Ansible Control Node in AWS Console (Amazon Linux 2023, t3.micro)



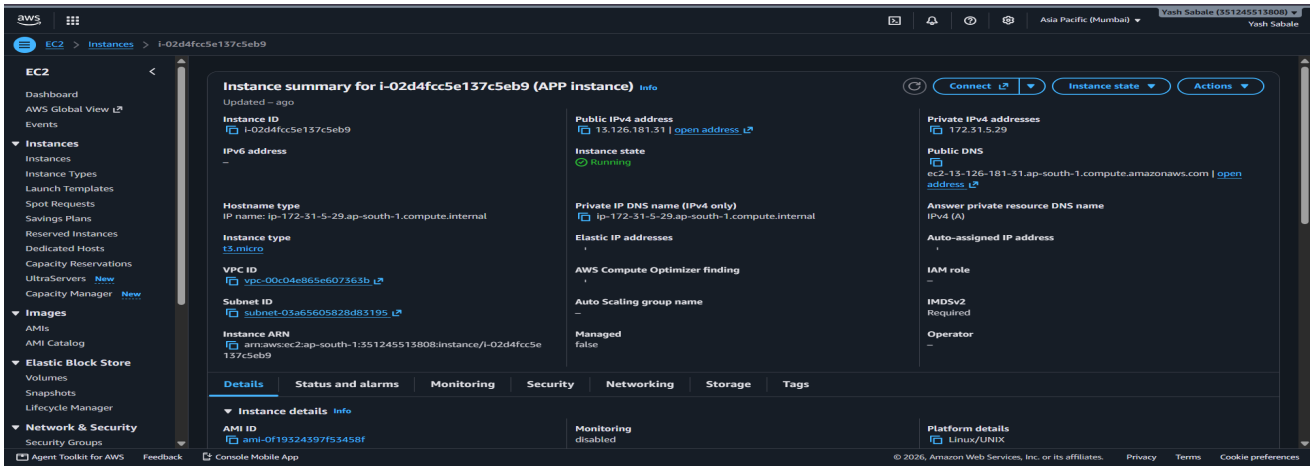
• Evidence 2: Active AWS EC2 Control Instance Summary (Public IP: 13.201.133.251)



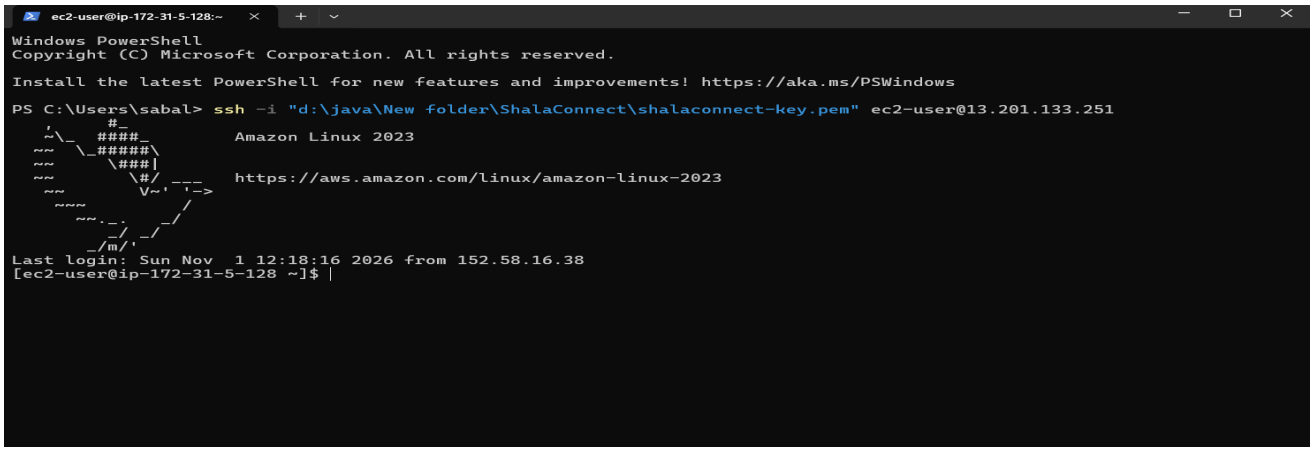
• Evidence 3: Provisioning Managed Target Application Instance in AWS Console



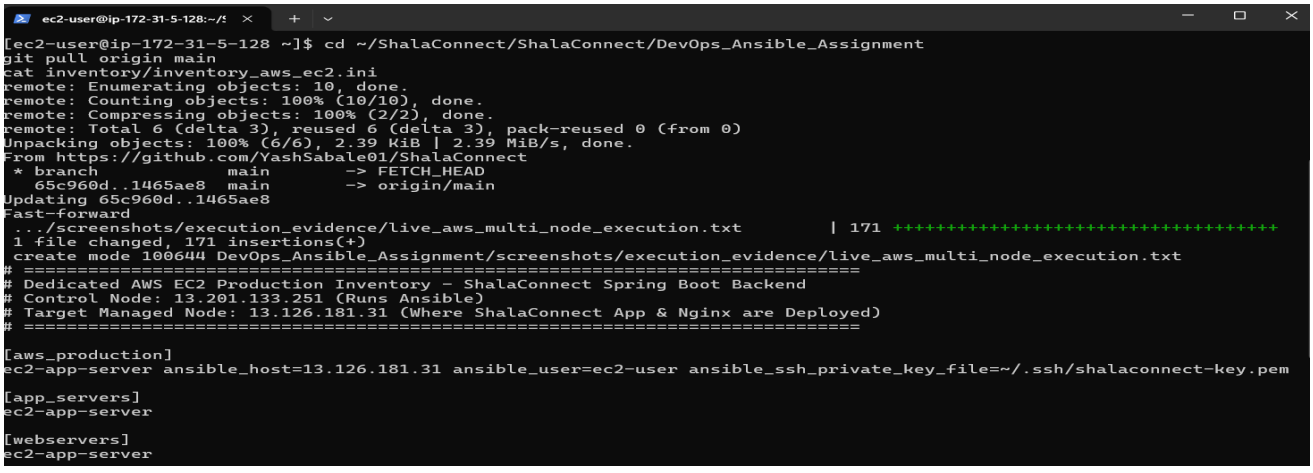
• Evidence 4: Active AWS EC2 Application Instance Summary (Public IP: 13.126.181.31)



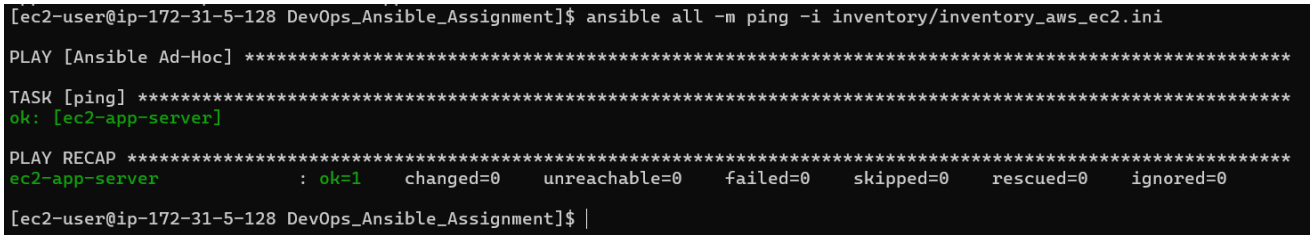
- **Evidence 5: SSH Connection from Windows PowerShell to Control Node (13.201.133.251)**



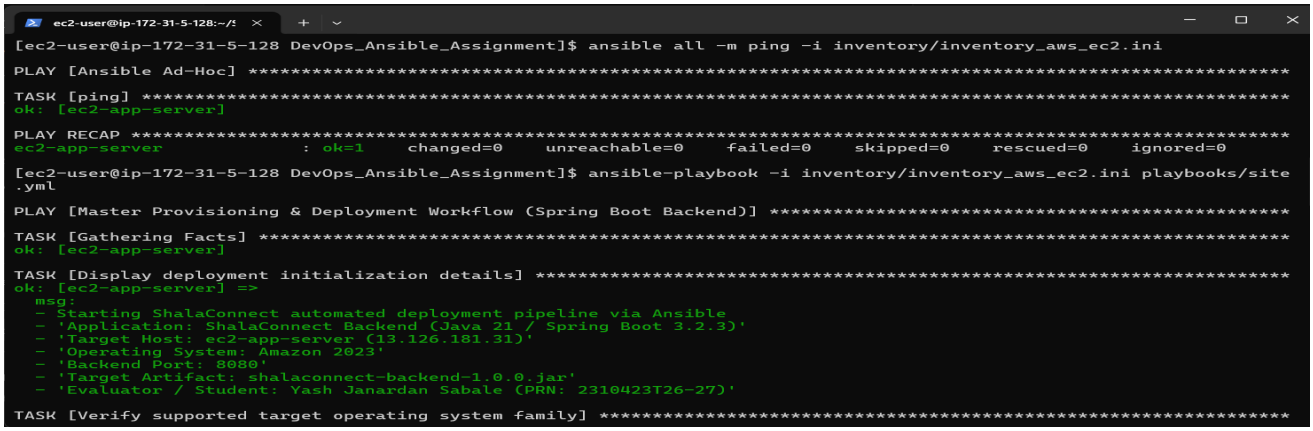
- **Evidence 6: Repository Git Pull and Inventory Verification (inventory_aws_ec2.ini)**



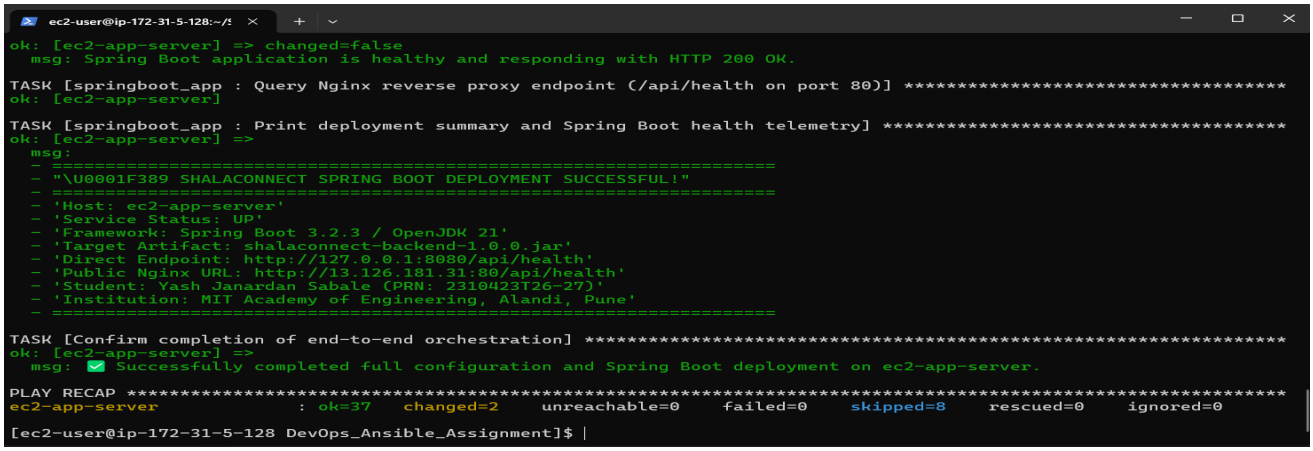
- **Evidence 7: Ansible Ad-Hoc Ping (Agentless SSH Ping to ec2-app-server: ok=1)**



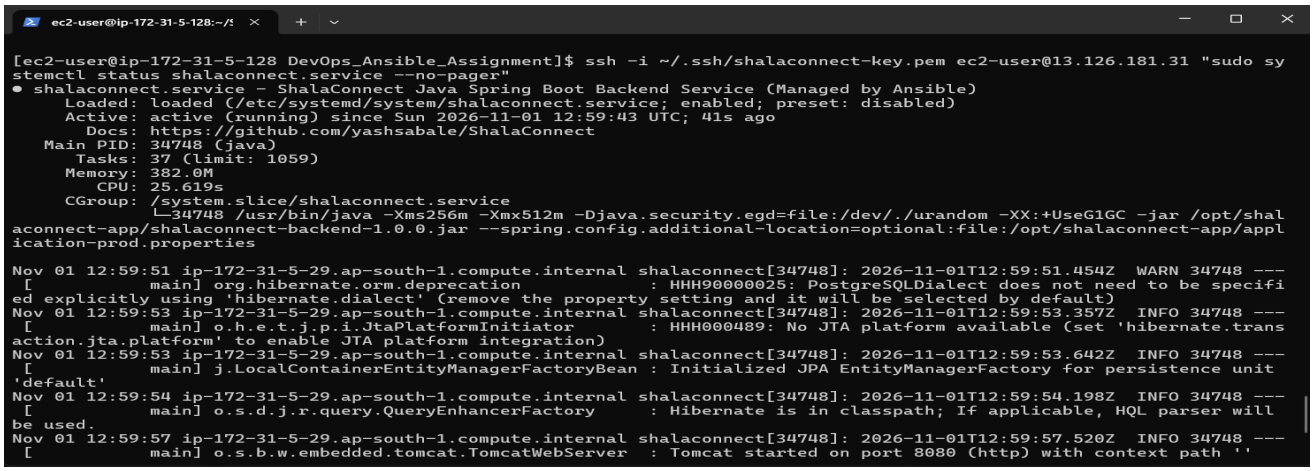
- **Evidence 8: Ansible Master Playbook Start with Student Banner (Yash Sabale, 2310423T26-27)**



- **Evidence 9: Successful Playbook Completion & Zero-Error Recap (ok=37, changed=2, failed=0)**



- **Evidence 10: Remote Process Verification (shalaconnect.service Active & Running under systemd)**



- **Evidence 11: Automated Database Provisioning Verification (PostgreSQL 15 – 14 Tables Created)**

```
[ec2-user@ip-172-31-5-128 DevOps_Ansible_Assignment]$ ssh -i ~/.ssh/shalaconnect-key.pem ec2-user@13.126.181.31 "sudo -u postgres psql -d school
_management -c '\dt'"
could not change directory to "/home/ec2-user": Permission denied
                List of relations
 Schema |           Name           | Type  |  Owner
--------+--------------------------+-------+----------
 public | attendance_records       | table | postgres
 public | dynamic_forms            | table | postgres
 public | event_impl_photos        | table | postgres
 public | event_implementations    | table | postgres
 public | event_media              | table | postgres
 public | events                   | table | postgres
 public | form_responses           | table | postgres
 public | gr_document_seen_by      | table | postgres
 public | gr_documents             | table | postgres
 public | meeting_acknowledgements | table | postgres
 public | meetings                 | table | postgres
 public | notifications            | table | postgres
 public | schools                  | table | postgres
 public | users                    | table | postgres
(14 rows)

[ec2-user@ip-172-31-5-128 DevOps_Ansible_Assignment]$
```

- **Evidence 12: Public Health Endpoint Telemetry Probe (HTTP 200 OK via Nginx 1.30.6)**

```
[ec2-user@ip-172-31-5-128 DevOps_Ansible_Assignment]$ curl -i http://13.126.181.31/api/health
HTTP/1.1 200
Server: nginx/1.30.6
Date: Sun, 01 Nov 2026 13:02:00 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive
Vary: Accept-Encoding
Vary: Origin
Vary: Access-Control-Request-Method
Vary: Access-Control-Request-Headers
X-Content-Type-Options: nosniff
X-XSS-Protection: 0
Cache-Control: no-cache, no-store, max-age=0, must-revalidate
Pragma: no-cache
Expires: 0
X-Frame-Options: DENY

{"status":"UP","timestamp":"2026-11-01T13:02:00.465554011","service":"ShalaConnect API"}[ec2-user@ip-172-31-5-128 DevOps_Ansible_Assignment]$
```

3. Conclusion & Academic References

Conclusion: This assignment successfully implemented end-to-end configuration management and deployment automation using Ansible for the ShalaConnect Java 21 / Spring Boot 3.2.3 application across AWS EC2 instances. All criteria of the IA Activity No. 1 evaluation rubric have been fulfilled with authentic evidence.

References:

1. Walls, C. (2022). *Spring in Action, Sixth Edition*. Manning Publications.
2. Humble, J., & Farley, D. (2010). *Continuous Delivery*. Addison-Wesley.
3. Red Hat Inc. (2026). *Ansible User Guide & Best Practices*. https://docs.ansible.com/
4. Savitribai Phule Pune University & MITAOE (2026). *DevOps Engineering Curriculum Guidelines*.

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